Fake universities as an emerging issue

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Fake universities are websites which claim to be reputable universities or part of the Academic Institutions. Indeed, they have no any relation with genuine universities, they are only websites created by cybercriminals. These "Universities" send spam emails to huge numbers of people, internationally, trying to cheat them. Fake universities claim to be accredited by a governmental or international institute and use names similar to reputable and known Universities. Physical addresses of fake universities are unknown or false and communication with them are only available by telephone number or email. Moreover, they show illustrations of their degree certificates in their websites to attract people [1].

Some fake universities also sell fake diplomas where applicants can select their desired degree, grade, major and etc [2]. These fake universities have negative effect on economic development and social cohesion and it is necessary to increase awareness of people about them [3]. Obviously, fake universities have financial goals, and with their behavior they produce destructive impacts on Science and sometimes it will be hard to detect what diploma or certificate are real. People who got their academic degrees from fake universities can apply for editorial members of journals. As they don't have knowledge and experiment in related domains, cannot do fair peer review, so number of false published papers in related journals will be increase.

Fake universities may be used for another purpose. They can gather valuable information from applicants in order to perform an identity theft. This application of fake universities is similar to a phishing attack. In information security literature, a phishing attack is an attempt for stealing users' sensitive information by using a fake website [4]. As universities need enrolment, requirements and information from applicant, they can gather list of valuable information and use them for identity theft. Also, patsy people will not suspect, because they think that are entering their information in a genuine university for an enrollment process. If we want to summarize which is the main purpose of cybercriminals for launching fake universities, financial purposes are their goals in side of gathering information.

At present, there are no exact statistics concerning the number of fake universities, but we can state that are decades. In effect, in September 2015, UGC (University Grants Commission) published a list of 23 fake universities in India [5], in May 2004, a report was published: 463 employees of the Indian federal government have fake academic degree [6].

To prevent damages and negative effects produced by fake universities, some Institutes (such as UCG) [7] report periodically a list of fake universities. Also, we can use IT security techniques to detect fake university. In general, many universities' websites have special domains such as ".edu" and ".ac.*", thus applicants can understand that they are facing with a genuine website. For example, applicants can trust to websites such as https://duke.edu or http://www.kcl.ac.uk, because these websites use academic domain which contain ".edu" or ".ac.uk". Cybercriminals cannot register those domains, because of any people who want to register such domains must have a license from science ministry or, however, from certified organizations. In some countries, such as Italy or Germany, universities don't use mentioned

domains and it is hard for applicants to verify their target university. In such case, we can suggest to applicants to use a simple approach to verify its target university. Fake universities are registered in recent years, so they have a lower number of visitors than genuine universities and they cannot receive a high page rank. Thus, we can check their domain page rank (http://checkpagerank.net/ http://www.whatsmypr.net) and WHOIS information (http://whois.domaintools.com) to detect them. For example, authentic website of *Bhundelkhand University* is www.bujhansi.org, a fake university will use www.bujhansi.info to cheat applicants. Page rank of authentic website is 5 and had been registered in 2007. Fake website have no any page rank (PR is equal to zero) and it appears to be registered in 2011. Table 1 shows more details for detecting an authentic university website from a fake one.

Table 1. Details for detecting an authentic university website from a fake one

No	Feature	Description	Authentic Website (www.bujhansi.org)	Fake Website (www.bujhansi.info)
1	Google Page Rank	Google page rank shows popularity of website and their ranking in Google search engine. Generally, authentic websites have more ranking than fake ones. Google page ranking is a number between 0 to 10. We use http://checkpagerank.net to find page rank of domains.	5	0
2	Alexa Rank	Alexa is a data base which shows ranking of a website with top visitors' countries. By using Alexa, we can find that visitors of a website come from which countries. We enter URL of universities in http://www.alexa.com and inspect return result.	Most visitors of this website (about 99.3%) come from India. Total websites which link this site are about 190.	This website has not more visitors. Total websites which link this site are about 12.
3	WHOIS Information	WHOIS database contain historical data about website such as domain registration time, screen shot history, registrar name and etc. We search URL of website in http://whois.domaintools.com and inspect result.	Creation date: 2007-01-09 Registrar: Dr Deepak Tomar registrant Organization: Bundelkhand University	Creation date: 2011-02-26 Registrar: Peter registrant Organization: Restricted

Fake universities don't have a Page Rank or have a low page rank. Their website had been registered in recent years by unknown people and organization. When we use Alexa database, we will find that their website traffic data is unavailable, because they have low number of visitors. In table 1, we tried to find original website of Bhundelkhand University. According to our inspection in this table, www.bujhansi.org is authentic website of this university. Alos, in UCG mentioned URL, had been introduced as the authentic website of this university (http://www.ugc.ac.in/stateuniversitylist.aspx?id=33&Unitype=2). There are many fake websites for Bhundelkhand University. In the authentic website of this university had been wrote that:

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"Disclaimer: It is to certify that bujhansi.org & bujhansi.ac.in are the official websites of Bundelkhand Univerity Jhansi, rest of the websites such as bujhansiresult.com, www.bundelkhanduniversity.org.in, bundelkhanduniversity.co.in, bujhansi.ind.in, www.bujhansiresult.org, www.bujhansi.info, www.bujhansi.net are all fake website and Bundelkhand University do not take responsibility for any of the content uploaded on these website".

It seems that fake universities are emerging as the issue for academic world and it is necessary to increase awareness of researchers and student about them. The full supplementary data of this research is available at http://academika.ir.

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